

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Winter 2023 Course: B. Tech. Branch: Mechanical Engineering/Production Engineering/Automobile Engineering Semester :VI Subject Code & Name: BTMPE603A_Y22 - IC ENGINE Max Marks: 60 Date:20/01/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/Co)	Marks
Q. 1	Solve Any Two of the following.		12
	a) Discuss about the working of a 4 stroke Diesel engine with neat sketch	Analyzing	06
	b) Explain about the various parts of an I.C. Engine with neat sketch.	Remembering	06
	c) Explain various classification of an I.C.Engines.	Analyzing	06
Q. 2	Solve Any Two of the following.		12
	a) Explain briefly the various phases of CI engine combustion.	Understanding	06
	b) Briefly explain the factors affecting normal combustion in SI engine.	Remembering	06
	c) Explain the phenomenon of Knock in Compression Ignition Engine.	Evaluating	06
Q. 3	Solve Any Two of the following.		12
	a) Explain Morse test and William's line method. What for they used?	Understanding & Remembering	06
	b) Why lubrication is required in the IC Engine. Explain with neat sketch Wet sump Lubrication system	Evaluating	06
	c) Explain the battery ignition system of engine with a neat sketch.	Analyzing	06
Q. 4	Solve Any Two of the following.		12
	a) State different exhaust emissions of I.C. Engine. Explain its effects on ecology.	Analyzing	06
	b) What are alternatives fuels and give the benefits of using Ethanol as an alternative fuel.	Evaluating	06
	c) What is a fuel cell? and how does it works?	Understanding & Evaluating	06

Q. 5	Solve Any Two of the following.		12
	a) What are the main components of electric vehicle	Understanding	06
	b) Explain briefly the components of Control system in IC Engine.	Evaluating	06
	c) Give advantages and disadvantages of Hybrid vehicles	Analyzing	06
	*** End ***		